Current: charge flow rate

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Learning target: Calculate current as charge transferred per time.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 2; 3 course days later.*
Force magnitude becomes 3 times as large, from Coulomb’s law.

02. Explain a worked example (worked)
Same steady series current through each section; energy is transferred, charge not used up.

03. Look and explain (model)
Conventional current follows that loop direction; steady current stops when the path opens.

04. Build the solution (practice)
3 A.

05. Use the idea (reason)
3 A; charge does not steadily accumulate at the junction.

06. Choose a method (mixed) *Review: Unit 6, Day 11; 12 course days later.*
Observed frequency is higher than 600 Hz because arriving wavefronts are closer together in time.

07. Exit check (exit)
6 C.

08. STRETCH: Apply it another way (stretch)
Examples 4 C/2 s and 10 C/5 s; current is rate.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 2; 3 course days later.*
$F_f/F_i=9/25\approx0.36$.

02. Explain a worked example (worked)
Same steady series current through each section; energy is transferred, charge not used up.

03. Represent and interpret (model)
Conventional current follows that loop direction; steady current stops when the path opens.

04. Solve and justify (practice)
$q=It=.25\times480=120$ C.

05. Analyze the reasoning (reason)
3 A; charge does not steadily accumulate at the junction.

06. Choose a method (mixed) *Review: Unit 6, Day 11; 12 course days later.*
$f_{obs}\approx637.5$ Hz, higher than emitted frequency.

07. Exit check (exit)
No; the circuit’s electromagnetic response propagates through the circuit while local charges drift much more slowly.

08. STRETCH: Apply it another way (stretch)
Triangle 6 C plus rectangle 8 C=14 C.